Nonlocal Biological Coherence: Coherence Operators Applied to Inter-Subject Correlation in Separated Dyads

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"The most beautiful thing we can experience is the mysterious. It is the source of all true art and all science. He to whom this emotion is a stranger, who can no longer pause to wonder and stand rapt in awe, is as good as dead: his eyes are closed."

— Albert Einstein, *The World As I See It* (1931)

This paper proposes measuring the mysterious with calibrated instruments.

Abstract

A small but persistent effect has been measured across decades of experiments: when two people interact and are then separated into electromagnetically shielded environments, stimulation of one subject produces measurable physiological correlates in the other. The meta-analytic effect size is small — $d = 0.11$ across 36 studies [Schmidt et al., 2004] — but it has survived multiple independent replications, registered reports, and publication bias analyses. Despite this, the phenomenon lacks a neutral measurement framework: existing analyses test for binary events (did the EEG spike correlate?) rather than measuring the continuous dynamics of inter-subject coupling. We propose applying the Coherence Engine operators [Thorarinson et al., 2026b] — validated on seven domains including turbofan degradation (100% detection rate, 185-cycle lead time) and cardiac arrhythmia prediction — to dual-subject EEG/HRV/EDA data from separated-dyad protocols. The operators provide a domain-agnostic vocabulary for measuring coupling dynamics: coherence rise during interaction, decay rate after separation, perturbation response during stimulation, and recovery to baseline. We present a pre-registered experimental protocol ($N = 60$ dyads, 64-channel EEG, Faraday cage isolation, double-blind stimulation) with explicit predictions for both positive and null outcomes. If the effect is real, the operators should detect temporal structure that point-event analyses miss. If the effect is null, the operators provide a more sensitive null result than binary correlation. Either way, we gain measurement precision on a question that has been asked for forty years but never with this class of instrument.

Keywords: coherence operators; inter-subject correlation; separated dyads; EEG; nonlocal correlation; measurement framework; pre-registered protocol; information geometry

1 Introduction: The Measurement Problem

There exists a body of peer-reviewed literature documenting correlated physiological responses between spatially separated human subjects. The corpus spans forty years, multiple laboratories,

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multiple neuroimaging modalities (EEG, fMRI, EDA), and has accumulated sufficient studies for quantitative meta-analysis. The effect is small. It is controversial. And it currently has no measurement framework adequate to its temporal structure.

The existing experimental paradigm — pioneered by Grinberg-Zylberbaum et al. [1994] and refined by Wackermann et al. [2003] — follows a consistent protocol: two subjects interact, are separated into shielded environments, one is stimulated, and the other's physiological response is measured. The analysis asks a binary question: *did the non-stimulated subject's EEG show a response correlated with the stimulation event?* This is analogous to measuring engine health by asking "did the engine fail today?" rather than tracking the continuous dynamics of degradation.

We propose a different approach. The Coherence Engine [Thorarinson et al., 2026b] provides six operators — persistence ($\mathcal{P}$), amplitude ($\mathcal{A}$), recovery ($\mathcal{R}$), deviation ($\mathcal{D}$), noise structure ($\mathcal{N}$), and inter-channel coherence (Δ) — that measure how a system's internal coordination changes over time. These operators have been validated on seven domains: turbofan engines (100% detection rate, 185-cycle lead time before failure), industrial valves, cardiac rhythm ($p < 10^{-177}$ for pre-arrhythmia detection), household energy, EEG seizure prediction (567-second mean lead time), financial fraud, and synthetic systems [Thorarinson et al., 2026a]. The operators are domain-agnostic: they measure structural properties of multivariate time series without requiring domain-specific knowledge.

The separated-dyad protocol has a natural temporal structure that maps directly onto coherence operator dynamics:

(i) **Coupling phase**: Subjects interact. If they establish physiological rapport, $\Delta(\mathrm{A}, \mathrm{B})$ should increase.

(ii) **Separation phase**: Subjects move to isolated environments. Δ should begin to decay — the rate of decay is itself informative.

(iii) **Stimulation phase**: Subject A receives stimuli. If nonlocal correlation exists, $\Delta(\mathrm{A}, \mathrm{B})$ should show perturbation-locked modulation during stimulus events.

(iv) **Recovery phase**: Post-stimulus, Δ should return to the inter-stimulus baseline — the recovery time constant $\mathcal{R}$ characterizes the system's return dynamics.

This is a richer measurement than "did subject B's EEG spike?" It asks: what are the *dynamics* of inter-subject coherence, and do they follow the same mathematical patterns observed when two coupled systems in other domains are perturbed?

We are not claiming telepathy. We are proposing to measure coupling dynamics between separated biological systems using operators that have been validated on systems where the coupling mechanism is known. If the operators detect nothing, that is an informative null result. If they detect structure, that structure can be characterized, quantified, and independently replicated.

2 Prior Evidence: A Tiered Assessment

We review the existing literature in three tiers, ranked by methodological rigor and replication status. For each study, we report the exact citation, effect size where available, sample size, methodological strengths and limitations, and replication status. This tiering is essential: the field has been damaged by treating all studies as equally evidential.

2.1 Tier 1: Strong Evidence

Tier 1 studies are peer-reviewed, published in indexed journals, and either meta-analytic in scope or independently replicated.

Wackermann et al. (2003). Wackermann et al. [2003] recorded six-channel EEG simultaneously from pairs of subjects in two acoustically and electromagnetically shielded rooms. Visual pattern-reversal stimuli were delivered to one subject; the other relaxed without stimulation. The effective voltage ratio Q of the averaged EEG signal at stimulus onset was compared to a reference distribution derived from non-stimulation periods. Significant departures from the reference distribution were found in most non-stimulated subjects ($N = 38$ subjects, 17 pairs). Published in *Neuroscience Letters* (Vol. 336, pp. 60–64).

Strengths: Double electromagnetic shielding (Faraday cages), quantitative EEG analysis, adequate sample size.
Limitations: Effect sizes not reported in standard units (Cohen's d); the Q-ratio metric is study-specific.
Replication: Partially replicated by Standish et al. [2004]; results contested in published commentary.

Standish et al. (2003, 2004). Two studies from the same laboratory. Standish et al. [2003] reported correlated fMRI signals between distant subjects ($N = 5$ pairs, $p < 0.05$). Standish et al. [2004] found EEG correlations between spatially and sensory isolated subjects ($N = 60$ sessions, published in *Journal of Alternative and Complementary Medicine*, Vol. 10, pp. 307–314).

Strengths: Both EEG and fMRI modalities tested; sensory isolation verified.
Limitations: Small fMRI sample ($N = 5$); single laboratory.
Replication: Internal replication across modalities; no independent replication.

Schmidt et al. (2004) — Meta-analysis. Schmidt et al. [2004] conducted two meta-analyses published in the *British Journal of Psychology* (Vol. 95, pp. 235–247). The Direct Mental Interaction in Living Systems (DMILS) meta-analysis combined 36 studies and found $d = 0.11$ ($p = .001$). A best-evidence synthesis of 7 high-quality studies yielded $d = 0.05$ ($p = .50$).

Strengths: Published in a mainstream psychology journal; systematic inclusion criteria; effect size reported in standard units.
Limitations: The best-evidence synthesis did not reach significance; substantial heterogeneity across studies.
Implication: The overall effect is small but non-zero. The best-evidence subset is inconclusive.

Tressoldi & Storm (2024) — Registered Report. Tressoldi and Storm [2024] published a Stage 2 Registered Report in *F1000Research* (PMC11134153) analyzing anomalous perception under Ganzfeld conditions across studies from 1974–2020. The overall effect size was ES = 0.08 (95% CI: 0.04–0.12), passing four publication bias tests. Selected participants showed effect sizes approximately three times larger than non-selected participants.

Strengths: Pre-registered analysis plan; passed Trim-and-Fill, PET-PEESE, p-curve, and selection model tests for publication bias.
Limitations: Ganzfeld is a perceptual task, not a physiological correlation measure; effect size is small.
Relevance: Demonstrates that anomalous information transfer effects survive rigorous bias testing.

Radin (2004) — Presentiment. Radin [2004] reported that electrodermal activity (EDA) was significantly higher before emotional photographs than before calm photographs ($p = 0.001$, $N = 133$ participants, 4,569 trials). The weighted mean per-trial effect size was $e = 0.064 \pm 0.015$, over 4 standard errors from null.

Strengths: Double-blind; three independent replications with new equipment, software, and participants; large trial count.
Limitations: The effect is temporal (pre-stimulus), not inter-subject; mechanism unknown.

Relevance: Demonstrates that biological systems show measurable anticipatory responses to future stimuli — a time-reversed coherence signature.

2.2 Tier 2: Moderate Evidence

Tier 2 studies are peer-reviewed but have limited or no independent replication.

Grinberg-Zylberbaum et al. (1994). The seminal "transferred potential" study [Grinberg-Zylberbaum et al., 1994]. Pairs of subjects interacted for 20 minutes, then were separated into Faraday chambers 14.5 m apart. When Subject A was stimulated with 100 light flashes and showed distinct evoked potentials, Subject B showed "transferred potentials" with similar morphology in approximately 25% of pairs. Control pairs (no prior interaction) showed no transferred potentials. Published in *Physics Essays* (Vol. 7, pp. 422–428).

Strengths: Faraday cage isolation; clear control condition; pioneering experimental design.
Limitations: Small sample; published in a low-impact journal; the 25% replication rate was never fully explained; the lead author disappeared under mysterious circumstances in 1994, preventing follow-up.
Replication: Partially replicated by Wackermann et al. [2003] and Standish et al. [2004]; never fully replicated with the same protocol.

Persinger et al. (2010). Persinger et al. [2010] exposed pairs of subjects in separate rooms to yoked circumcerebral rotating magnetic fields while one subject viewed flashing lights at various frequencies. Reliable increases in QEEG power at corresponding frequencies were observed in the non-stimulated subject's right parietal region, but only when the shared magnetic field temporal pattern matched the duration of normal cortical microstates ($\sim$100 ms). Published in *Neuroscience Letters* (Vol. 486, pp. 231–234).

Strengths: Specific frequency-matching predictions confirmed; published in a mainstream neuroscience journal.
Limitations: Circumcerebral magnetic stimulation is a non-standard intervention; small sample sizes; Persinger's later work on "God helmet" effects was contested.
Replication: Partially replicated across the Atlantic Ocean by the same group; no independent replication.

Achterberg et al. (2005). Achterberg et al. [2005] used fMRI to measure brain activity in subjects whose paired healers sent "distant intention" at random 2-minute intervals unknown to the recipient. Significant activation was observed in anterior and middle cingulate, precuneus, and related regions during send vs. no-send periods ($p = .0001$, $N = 11$ healer-recipient pairs). Published in *Journal of Alternative and Complementary Medicine* (Vol. 11, pp. 965–971).

Strengths: fMRI provides spatial specificity; double-blind timing; large effect.
Limitations: Small N; self-selected healer-recipient pairs; no stranger-control condition.
Replication: Not independently replicated.

McCraty et al. (2009). McCraty et al. [2009] reviewed HeartMath Institute research on heart rate variability (HRV) coherence, including studies of physiological coherence between individuals at distance. Published in *Integral Review* (Vol. 5, No. 2).

Strengths: Institutional research program spanning two decades; HRV is a well-characterized physiological measure.
Limitations: Published in a non-mainstream journal; some studies lack independent replication; institutional bias concerns.
Replication: Internal replications only.

2.3 Tier 3: Suggestive Evidence

Tier 3 studies have methodological concerns, are contested, or lack peer review in mainstream journals.

Targ & Puthoff (1974). Targ and Puthoff [1974] published "Information transmission under conditions of sensory shielding" in *Nature* (Vol. 251, pp. 602–607), reporting remote viewing results from SRI International. The CIA's Stargate program subsequently funded ∼\$20 million in remote viewing research over two decades.

Strengths: Published in *Nature*; replicated internally at SRI; declassified CIA evaluation concluded "a statistically significant effect."
Limitations: *Nature*'s editorial board published the paper with reservations about experimental oversight; the final AIR evaluation (1995) concluded the effect was not useful for intelligence operations; methodological criticisms by Marks, Hyman, and others.

Sheldrake (2003). Sheldrake and Smart [2003] reported above-chance identification of telephone callers ($N = 63$ participants, 571 trials, hit rate 42% vs. 25% expected, $p = 4 \times 10^{-16}$). Published in the *Journal of the Society for Psychical Research.*

Strengths: Simple protocol; large effect size; randomized selection of callers.
Limitations: Published in a parapsychology journal; potential for subtle sensory cues; incomplete documentation of randomization procedures.

PEAR Laboratory (1979–2007). Jahn et al. [1997] reported a 12-year program at Princeton Engineering Anomalies Research (PEAR) lab finding small but consistent correlations between human intention and random event generator output. The effect was very small ($z \approx 0.7$ per run).

Strengths: Large database; consistent methodology over 12 years.
Limitations: Very small effect; the Bösch et al. meta-analysis of RNG studies [Bösch et al., 2006] found that the small effect could be explained by publication bias; PEAR lab closed in 2007 without independent replication of its full program.

2.4 Synthesis

Figure 4 summarizes the evidence pyramid. The Tier 1 evidence establishes that: (a) a small but statistically significant effect has been measured ($d = 0.11$, Schmidt et al.); (b) this effect survives publication bias testing (Tressoldi & Storm); and (c) anticipatory physiological responses to future stimuli are measurable (Radin). The Tier 2 evidence provides specific mechanistic suggestions (frequency-matching, regional brain activation) that a measurement framework should be able to detect. The Tier 3 evidence is not relied upon for any claim in this paper.

What is missing from all tiers is a continuous measurement of the *dynamics* of inter-subject coupling — how it develops, how it decays, how it responds to perturbation, and how it recovers. This is what the coherence operators provide.

3 The Coherence Framework

We provide a brief summary of the operators relevant to inter-subject correlation measurement. Full derivations appear in Thorarinson et al. [2026b]; validation benchmarks across seven domains appear in Thorarinson et al. [2026a].

3.1 Operator Definitions

The Coherence Engine operates on a multivariate time series $\mathbf{x}(t) \in \mathbb{R}^n$ observed over a sliding window $[t-w, t]$. Six operators extract structural properties of the time series:

Definition 1 (Persistence $\mathcal{P}$). *Persistence measures the autocorrelation structure of the system. For a time series $x(t)$, the persistence operator is the detrended fluctuation analysis (DFA) scaling exponent α:*

$$\mathcal{P}[\mathbf{x}](t) = \alpha(t) \quad \textit{where} \quad F(s) \sim s^{\alpha} \tag{1}$$

For a healthy coupled system, $\alpha \approx 1.0$ (long-range correlation); for a decoupled system, $\alpha \to 0.5$ (uncorrelated random walk).

Definition 2 (Amplitude $\mathcal{A}$). *Amplitude measures the magnitude of the system's variability relative to its historical baseline:*

$$\mathcal{A}[\mathbf{x}](t) = \frac{\sigma_w(t)}{\sigma_{baseline}} \tag{2}$$

where $\sigma_w(t)$ is the standard deviation in the current window and $\sigma_{baseline}$ is the historical reference.

Definition 3 (Recovery $\mathcal{R}$). *Recovery measures the characteristic time for the system to return to baseline after perturbation:*

$$\mathcal{R}[\mathbf{x}](t) = \tau_{1/2} \quad \textit{where} \quad |\mathbf{x}(t+\tau_{1/2}) - \bar{\mathbf{x}}_0| = \frac{1}{2}|\mathbf{x}(t_p) - \bar{\mathbf{x}}_0| \tag{3}$$

and t_p is the perturbation time and $\bar{\mathbf{x}}_0$ is the pre-perturbation baseline.

Definition 4 (Deviation $\mathcal{D}$). *Deviation measures the distance of the system's current operating point from its historical baseline manifold:*

$$\mathcal{D}[\mathbf{x}](t) = d_{FR}\big(p_w(t),\ p_0\big) \tag{4}$$

where d_{FR} is the Fisher-Rao distance between the current window's empirical distribution $p_w(t)$ and the baseline distribution p_0.

Definition 5 (Noise Structure $\mathcal{N}$). *Noise structure measures the spectral composition of the system's variability:*

$$\mathcal{N}[\mathbf{x}](t) = \beta(t) \quad \textit{where} \quad S(f) \sim f^{-\beta} \tag{5}$$

For a healthy system, $\beta \approx 1$ (pink noise); degradation produces $\beta \to 0$ (white noise) or $\beta \to 2$ (Brownian noise, indicating loss of regulation).

Definition 6 (Inter-Channel Coherence Δ). *The central operator for inter-subject measurement. Given two multivariate time series $\mathbf{x}_A(t)$ and $\mathbf{x}_B(t)$ from subjects A and B, the inter-subject coherence is:*

$$\Delta[\mathbf{x}_A, \mathbf{x}_B](t) = \frac{1}{|\mathcal{F}|} \sum_{f \in \mathcal{F}} \frac{|S_{AB}(f,t)|^2}{S_{AA}(f,t) \cdot S_{BB}(f,t)} \tag{6}$$

where $S_{AB}(f,t)$ is the cross-spectral density between A and B at frequency f in the current window, and $\mathcal{F}$ is the set of frequency bands of interest (delta, theta, alpha, beta, gamma for EEG).

3.2 Mapping to the Separated-Dyad Protocol

The coherence operators map naturally to the four phases of the separated-dyad protocol (Figure 1):

Phase 1: Coupling. During face-to-face interaction, if the subjects establish physiological rapport, we predict:

- $\Delta(A, B)$ increases from baseline (chance-level coherence) toward a coupling plateau.
- $\mathcal{P}$ for the joint system $[\mathbf{x}_A; \mathbf{x}_B]$ increases, indicating the emergence of long-range correlation in the combined time series.
- $\mathcal{N}$ for the joint system shifts toward $\beta = 1$ (pink noise), consistent with the emergence of coordination.

Phase 2: Separation. After subjects are moved to isolated Faraday cages:

- If the coupling was real, $\Delta(A, B)$ should decay — but the *rate* of decay is informative. Exponential decay with time constant τ_{sep} characterizes the coupling's persistence after separation.
- If the coupling was an artifact of shared environment, Δ should drop immediately to chance levels.

Phase 3: Stimulation. When Subject A receives stimuli (light/sound):

- Subject A's individual operators ($\mathcal{A}_A$, $\mathcal{P}_A$) respond to the stimulus — this is the expected evoked response.
- The key measurement: does $\Delta(A, B)$ show stimulus-locked modulation? Specifically, does Δ increase during stimulation events relative to inter-stimulus intervals?
- $\mathcal{R}$ for Δ after each stimulus event measures how quickly the inter-subject coherence returns to baseline.

Phase 4: Decoupling Control. The same protocol with stranger pairings provides the critical comparison:

- $\Delta_{\text{bonded}}(A, B)$ vs. $\Delta_{\text{stranger}}(A, B)$ during stimulation events.
- The null hypothesis: $\Delta_{\text{bonded}} = \Delta_{\text{stranger}}$ for all phases.

3.3 Why This Is More Sensitive Than Binary Correlation

Existing analyses extract a single number from each session: "did Subject B's EEG show an event-related response correlated with Subject A's stimulation?" This discards the temporal dynamics entirely. The coherence operators preserve the full time course:

1. A binary correlation test has one opportunity to detect the effect per stimulus event. The coherence operator $\Delta(t)$ is a continuous function, providing temporal resolution equal to the sliding window step size.
2. Binary tests cannot distinguish between a brief, strong correlation and a sustained, weak correlation. The operators distinguish these via $\mathcal{A}$ (amplitude) and $\mathcal{P}$ (persistence).
3. Binary tests cannot detect coupling that decays gradually after separation. The operators measure the decay rate τ_{sep} directly.
4. Binary tests cannot detect anticipatory coherence shifts before stimulation events. The operators' lead-time capability (validated at 185 cycles on turbofan data and 567 seconds on EEG seizure data) can detect pre-stimulus coherence changes.

4 Proposed Experimental Protocol

This section describes a protocol designed for pre-registration on the Open Science Framework (OSF) before data collection. All analysis procedures, primary outcomes, and statistical tests are specified *a priori* [Nosek et al., 2018, Chambers, 2013].

4.1 Participants

- $N = 60$ dyads total: 30 bonded pairs (romantic partners, close siblings, long-term friends with ≥ 5 years of relationship) and 30 stranger-control pairs (matched for age, sex, and handedness).
- Stranger pairs are formed by re-pairing members of different bonded dyads, ensuring no prior relationship.
- Exclusion criteria: neurological disorders, psychoactive medication, prior meditation training > 100 hours (to control for a known moderator in the Ganzfeld literature).
- Power analysis: for the primary effect $d = 0.11$ [Schmidt et al., 2004], $\alpha = 0.01$ (Bonferroni-corrected for multiple comparisons across frequency bands), and power = 0.80, the required sample size is $N \approx 1,600$ per group for a two-sample t-test. This is infeasible. However, the coherence operator approach has two advantages: (a) each dyad contributes a continuous time series, not a single binary outcome, providing $\sim$30 stimulus events $\times$ 5 frequency bands = 150 measurements per dyad; (b) the operators may detect temporal structure invisible to binary tests, potentially yielding a larger effective effect size. We set $N = 60$ dyads as a pragmatic choice, acknowledge that this is underpowered for $d = 0.11$ on a per-event basis, and pre-register the sensitivity analysis: given $N = 60$ dyads and $\alpha = 0.01$, the minimum detectable effect is $d \approx 0.47$ for a two-sample t-test, or substantially smaller when exploiting the repeated-measures structure.

4.2 Physiological Measurements

1. **EEG**: 64-channel, 500 Hz sampling rate, 10–20 system with additional temporal and parietal coverage. System: BioSemi ActiveTwo or equivalent.
2. **Heart Rate Variability (HRV)**: Polar H10 chest strap, RR-interval resolution ≤ 1 ms. Provides the time-domain signal for autonomic coherence measurement.
3. **Electrodermal Activity (EDA)**: Skin conductance measured from non-dominant hand, 128 Hz sampling. Provides the sympathetic nervous system correlate used in prior DMILS studies.

All three modalities are recorded simultaneously and time-synchronized via hardware triggers.

4.3 Electromagnetic Shielding

- Each subject is placed in a separate Faraday cage with verified electromagnetic attenuation ≥ 80 dB at frequencies from 100 kHz to 10 GHz [Ott, 2009].
- Cages are in separate rooms, minimum 10 m apart, with independent HVAC and lighting circuits.
- EM monitoring: a spectrum analyzer continuously records the electromagnetic environment inside each cage throughout the experiment. Any session with anomalous EM activity is flagged and excluded from primary analysis.
- Acoustic isolation: ambient noise < 30 dB SPL in each cage; no shared acoustic pathways.

4.4 Protocol Timeline

See Figure 1 for the complete timeline.

1. **Baseline** (10 min): Both subjects in their respective Faraday cages, eyes closed, no task. Establishes individual physiological baselines and chance-level $\Delta(A, B)$.

2. **Coupling** (20 min): Subjects are brought to a shared room. They sit facing each other and perform a guided synchronized breathing exercise (4 s inhale, 4 s exhale, 4 s hold) for 10 minutes, followed by 10 minutes of free conversation/interaction. EEG/HRV/EDA recorded throughout.

3. **Separation** (5 min): Subjects return to their respective Faraday cages. Recording continues during transit and settling. The transition from shared-environment coherence to isolated coherence is a key measurement.

4. **Stimulation** (15 min): Subject A receives 30 stimuli at pseudo-random intervals (mean 30 s, range 15–60 s, uniform distribution). Stimuli are combined light (200 ms white flash, 100 cd/m^2) and sound (200 ms 1000 Hz tone, 70 dB SPL). Subject B remains in the dark, eyes closed, with white noise masking at 50 dB SPL.

   Double-blind: The experimenter monitoring Subject B does not know when stimuli are delivered to Subject A. Stimulus timing is controlled by a computer in a third room with no communication channel to Subject B's environment.

5. **Rest** (5 min): No stimuli. Both subjects rest. Measures post-stimulation recovery of Δ.

6. **Stranger control**: After a 15-minute break, each subject is re-paired with a stranger (a member of a different bonded dyad). Phases 1–5 are repeated with the stranger pair. This within-subjects design controls for individual differences in baseline physiology.

Total session duration: approximately 2.5 hours per subject.

4.5 Pre-Registered Analysis Plan

Primary outcome. The stimulus-locked change in inter-subject coherence:

$$\Delta\Delta_{\text{stim}} = \overline{\Delta}_{[-1,+3]\text{s}} - \overline{\Delta}_{[-5,-2]\text{s}} \tag{7}$$

where the windows are relative to stimulus onset. This is computed separately for bonded and stranger pairs. The primary test is:

$$H_0 : \Delta\Delta^{\text{bonded}}_{\text{stim}} = \Delta\Delta^{\text{stranger}}_{\text{stim}} \tag{8}$$

tested with a paired permutation test (10,000 permutations), Bonferroni-corrected across 5 EEG frequency bands (delta: 1–4 Hz, theta: 4–8 Hz, alpha: 8–13 Hz, beta: 13–30 Hz, gamma: 30–50 Hz). Significance threshold: $\alpha = 0.01$.

Secondary outcomes.
1. **Coupling decay rate**: Fit $\Delta(t)$ during the separation phase to $\Delta(t) = \Delta_0 \cdot e^{-t/\tau} + \Delta_\infty$. Compare τ_{bonded} vs. τ_{stranger}.
2. **Recovery time**: Post-stimulus $\mathcal{R}$ for $\Delta(A, B)$. Compare $\mathcal{R}_{\text{bonded}}$ vs. $\mathcal{R}_{\text{stranger}}$.
3. **Pre-stimulus anticipation**: Test whether Δ begins increasing before stimulus onset (within $[-3, 0]$ s window). This is motivated by the presentiment literature [Radin, 2004] and the operators' lead-time capability.
4. **Modality comparison**: Coherence computed separately for EEG, HRV, and EDA. Which physiological modality shows the largest inter-subject effect?
5. **Frequency band specificity**: Which EEG frequency band shows the largest $\Delta\Delta_{\text{stim}}$? Prior work suggests alpha (8–13 Hz) is the most relevant [Wackermann et al., 2003].

Exploratory analyses. Not pre-registered and reported as exploratory:
- Relationship between self-reported relationship closeness and Δ amplitude.
- Individual differences in baseline coherence as moderators.
- Cross-frequency coupling between Subject A's alpha and Subject B's theta.

5 Predicted Results

We present explicit predictions for both outcomes (Figure 2).

Prediction 1 (Positive Result)**.** *If nonlocal inter-subject correlation is real:*

(a) $\Delta\Delta_{stim}^{bonded} > \Delta\Delta_{stim}^{stranger}$ *($p < 0.01$, permutation test).*

(b) The coupling decay follows $\Delta(t) = \Delta_0 \cdot e^{-t/\tau} + \Delta_\infty$ *with* $\tau_{bonded} > \tau_{stranger}$ *— bonded pairs maintain coherence longer after separation.*

(c) Recovery time $\mathcal{R}_{bonded} < \mathcal{R}_{stranger}$ *— bonded pairs recover faster from stimulus perturbation, consistent with stronger coupling.*

(d) The effect is largest in the alpha band (8–13 Hz), based on prior literature.

(e) EDA may show larger effects than EEG, based on the DMILS literature which predominantly used EDA.

Prediction 2 (Null Result)**.** *If the effect does not exist:*

(a) $\Delta\Delta_{stim}^{bonded} \approx \Delta\Delta_{stim}^{stranger}$ *across all frequency bands and modalities.*

(b) Coupling decay rates are identical: $\tau_{bonded} = \tau_{stranger}$*.*

(c) No pre-stimulus anticipatory coherence changes.

(d) This null result is more informative than previous nulls because the operators measure continuous dynamics, not binary events. A null result with calibrated operators that detect 567-second lead times in EEG seizure prediction rules out coupling dynamics at that sensitivity level.

Both outcomes are scientifically valuable. A positive result provides the first continuous-dynamics characterization of inter-subject coupling. A null result with calibrated instruments provides a stronger constraint than any previous experiment.

6 Effect Size in Context

The effect size $d = 0.11$ reported by Schmidt et al. [2004] is small by any standard. However, small does not mean negligible. Figure 3 compares this effect to other established small effects in medicine and psychology:

- **Aspirin and myocardial infarction**: $r = 0.034$ ($d \approx 0.068$). The Physicians' Health Study [Steering Committee of the Physicians' Health Study Research Group, 1989] found a 44% reduction in MI risk, corresponding to an effect size so small that Rosenthal [1990] used it as the canonical example of a "small effect with large consequences." The study was terminated early on ethical grounds — despite $d \approx 0.07$.
- **Psychotherapy for depression**: $d = 0.22$ in high-quality studies [Cuijpers et al., 2010]. When study quality is controlled, the effect of psychotherapy is comparable to the DMILS effect.
- **Calcium supplementation and bone density**: $d = 0.31$ at lumbar spine [Tai et al., 2015]. Comparable to the effect size in selected-participant Ganzfeld studies.
- **DMILS inter-subject correlation**: $d = 0.11$ [Schmidt et al., 2004].
- **Ganzfeld anomalous perception**: ES = 0.08 [Tressoldi and Storm, 2024].

The point is not that these effects are equivalent in importance or mechanism. The point is that $d = 0.11$ is within the range of effects that medicine and psychology accept as real and act upon. The question is not whether $d = 0.11$ is "too small to matter" — the question is whether the measurement framework is adequate to characterize it.

7 Alternative Hypotheses

Every criticism of the inter-subject correlation literature is addressed by specific design features of the proposed protocol. We list each alternative hypothesis and its control.

1. Electromagnetic leakage. If Subject A's stimulus creates electromagnetic radiation that propagates to Subject B's environment, this could produce correlated signals without any nonlocal effect.

Control: Faraday cage shielding (≥ 80 dB attenuation). Continuous EM spectrum monitoring inside each cage. Any session with anomalous EM activity is excluded. Additionally: the stimulus (200 ms flash + tone) produces negligible EM radiation outside the cage, and the cage attenuation ensures that any residual signal is far below EEG amplifier noise floors.

2. Shared environment. If subjects share vibrational coupling through building structure, HVAC, or other environmental channels, this could produce correlated physiological responses.

Control: Separate rooms with independent HVAC and lighting. Minimum 10 m separation. The stranger-control condition (same rooms, same equipment, same environment) is the critical comparison: if the effect is environmental, Δ_{stranger} should equal Δ_{bonded}.

3. Statistical artifacts. Multiple comparisons, p-hacking, and post-hoc analysis selection could inflate Type I error rates.

Control: Pre-registered analysis plan on OSF. Bonferroni correction across frequency bands. Permutation tests (10,000 permutations) for the primary outcome, which are distribution-free and robust to non-normality [Good, 2005]. Exploratory analyses reported separately with appropriate caveats.

4. Publication bias. The literature may overestimate the true effect because null results are not published.

Control: We cannot eliminate publication bias in the existing literature. We can note that the Tressoldi & Storm registered report [Tressoldi and Storm, 2024] passed four independent publication bias tests. Our proposed study is pre-registered and will be published regardless of outcome.

5. Subtle sensory cues. If the experimenter monitoring Subject B has information about when stimuli occur, they could inadvertently provide cues through body language, breathing, or other subtle channels.

Control: Double-blind design. The experimenter with Subject B has no information about stimulus timing. Stimulus delivery is automated from a separate, isolated room. Subject B wears white noise masking. No human is present in Subject B's Faraday cage during the stimulation phase — all monitoring is via remote physiological recording.

6. Prior probability argument. The prior probability that nonlocal biological correlation exists, given current physics, is very low. Therefore, any positive result should be treated with extreme skepticism.

Response: This is a valid concern, and we take it seriously. Our response is threefold: (a) We are not claiming a mechanism. We are measuring a correlation using validated operators. (b) The meta-analytic evidence ($d = 0.11$ across 36 studies, $p = .001$) constitutes Bayesian evidence that should update even a skeptical prior. (c) The operators are domain-agnostic — they do not "know" whether they are measuring turbofan engines or human brains. If they detect structure, that structure must be explained; if they do not, the null is stronger than previous nulls.

7. "Extraordinary claims require extraordinary evidence." Nonlocal biological correlation, if real, would conflict with current physics. The evidence must therefore be held to a higher standard.

Response: We agree, which is why we propose pre-registration, double-blind design, Faraday cage isolation, permutation tests, and explicit predictions for both positive and null outcomes.

But we note that "extraordinary evidence" is a rhetorical standard, not a statistical one. The pre-registered significance threshold ($\alpha = 0.01$ with Bonferroni correction) is more conservative than most published neuroscience. And we note that aspirin's effect on myocardial infarction ($d \approx 0.07$) was considered extraordinary enough to terminate a clinical trial early on ethical grounds.

8 The Structural Hole

In a companion paper [Thorarinson, 2026], we demonstrated that concepts lacking single-word English equivalents occupy measurably isolated regions of English embedding space, and that this isolation produces a feedback loop: missing word → missing search query → missing retrieval → missing research → the concept remains unstudied.

The phenomenon investigated here is a case study of this feedback loop.

English has no neutral scientific word for correlated physiological responses between separated subjects. The available terms carry heavy connotative baggage:

- **Telepathy**: From the Greek $\tau\tilde{\eta}\lambda\epsilon$ (distant) + $\pi\acute{\alpha}\theta o\varsigma$ (feeling). Coined by Frederic Myers in 1882. Now connotes supernatural belief, X-Men, and Uri Geller. No serious scientist can use this word without career risk.
- **Psi**: A catch-all term from parapsychology that encompasses everything from remote viewing to psychokinesis. Too broad and too stigmatized.
- **Anomalous cognition**: A CIA euphemism from the Stargate program. Clinically neutral but bureaucratic.
- **Nonlocal correlation**: Borrowed from quantum mechanics, where it has a precise technical meaning that may or may not apply here.

Other languages have terms that are more neutrally descriptive:

- **Chinese**: 心灵感应 (xīnlíng gǎnyìng) — literally "heart-spirit mutual response." This describes the phenomenon without asserting a mechanism: two systems respond to each other.
- **Sanskrit**: *siddhi* — "accomplished power" or "attainment." In the Yoga Sutras of Patanjali, *siddhis* are described as natural consequences of systematic contemplative practice, not as supernatural gifts.
- **Japanese**: 以心伝心 (*ishin-denshin*) — "what the mind thinks, the heart transmits." A standard Japanese expression for wordless communication between people who know each other well.

The Chinese term is particularly apt: "heart-spirit mutual response" describes exactly what the coherence operators measure — the mutual response between two systems' internal dynamics. It does not claim a mechanism; it describes an observation.

The absence of a neutral English term is not merely a linguistic curiosity. It is a structural hole in the sense of Burt [1992]: a gap in the network of concepts available to English-language science. This gap has hindered investigation in two ways:

1. **Stigma by association**: Any researcher who studies "telepathy" faces career consequences, regardless of methodology. The word contaminates the research.
2. **Retrieval failure**: There is no PubMed MeSH heading, no grant category, and no departmental home for this research. It falls between neuroscience, psychology, and physics, belonging fully to none.

This paper uses "inter-subject coherence" throughout, because that is what the operators measure: coherence between subjects. The term is precise, neutral, and requires no metaphysical commitments.

9 Connection to Information Geometry

The coherence operators have a natural interpretation in information geometry [Amari, 2016, Rao, 1945]. This section develops the connection, which provides both mathematical grounding and a route to quantitative predictions.

9.1 The Fisher-Rao Manifold of Physiological States

Each subject's physiological state at time t is characterized by a probability distribution p_t over possible measurement outcomes (EEG amplitudes, RR intervals, skin conductance levels). The space of all such distributions forms a statistical manifold $\mathcal{M}$, equipped with the Fisher-Rao metric:

$$ds^2 = \sum_{i,j} g_{ij}(\theta)\, d\theta^i\, d\theta^j, \qquad g_{ij}(\theta) = \mathbb{E}_\theta \left[\frac{\partial \log p(x|\theta)}{\partial \theta^i} \cdot \frac{\partial \log p(x|\theta)}{\partial \theta^j} \right] \tag{9}$$

where θ parameterizes the distribution and g_{ij} is the Fisher information matrix [Fisher, 1925].

9.2 Coupling as Manifold Alignment

When two independent subjects are measured simultaneously, their joint state space is the product manifold $\mathcal{M}_A \times \mathcal{M}_B$. Their joint distribution is $p(x_A, x_B) = p_A(x_A) \cdot p_B(x_B)$ — a product distribution, indicating statistical independence. The joint state moves along the product manifold, and the inter-subject coherence Δ is at chance level.

If the subjects establish coupling (during the interaction phase), their joint distribution deviates from the product form:

$$p(x_A, x_B) \neq p_A(x_A) \cdot p_B(x_B) \tag{10}$$

Geometrically, the joint state leaves the product manifold and moves onto a *lower-dimensional submanifold* of the full joint space — a submanifold that encodes the coupling structure. The coherence operator Δ measures the distance from the product manifold:

$$\Delta(A, B) \propto d_{\mathrm{FR}}\big(p_{AB}(t),\ p_A(t) \otimes p_B(t)\big) \tag{11}$$

This is the Fisher-Rao distance between the observed joint distribution and the product of its marginals — a model-free measure of statistical dependence.

9.3 Predictions from the Geometric Framework

The geometric interpretation generates three quantitative predictions:

1. **Coupling increases during interaction**: The joint state moves off the product manifold. The dimensionality of the occupied submanifold decreases as coupling develops, measurable as a reduction in the effective rank of the joint covariance matrix.

2. **Separation causes geodesic drift**: After separation, if no nonlocal correlation persists, the joint state returns to the product manifold along a geodesic (shortest path in the Fisher-Rao metric). The decay rate τ_{sep} corresponds to the geodesic distance divided by the drift velocity.

3. **Stimulation causes tangent vector alignment**: If Subject A's response to stimulation produces a correlated response in Subject B (nonlocal coupling), then the tangent vectors to the two subjects' trajectories on $\mathcal{M}$ become aligned at the time of stimulation. This alignment is detectable as a transient increase in Δ.

The essential point is that this is the same geometry used by the coherence operators on engineering systems [Thorarinson et al., 2026b]. The Fisher-Rao metric does not care whether the systems are turbofan engines or human brains. It measures the geometry of probability distributions, and coupling between any two systems manifests as the same geometric signature: departure from the product manifold.

10 Limitations

This paper has significant limitations that we state explicitly.

1. **We have not run this experiment.** This is a protocol paper. The predictions are untested. All results are prospective.

2. **The prior probability is low.** No known physical mechanism supports nonlocal biological correlation at the macroscopic scale. Quantum entanglement, while formally nonlocal, has not been demonstrated to survive the thermal decoherence of biological systems at room temperature — though recent work in quantum biology [Engel et al., 2007, Lambert et al., 2013] has challenged assumptions about decoherence timescales.

3. **The effect size is small.** Even if real, $d = 0.11$ is not clinically significant. It is comparable to aspirin's effect on heart attacks — measurably real, worth studying, but not transformative for any individual.

4. **The study is underpowered for $d = 0.11$.** We have addressed this transparently in Section 4: the repeated-measures structure and continuous operator measurements may increase effective power, but we cannot guarantee adequate sensitivity without running the study.

5. **The coherence operators are not magic.** They measure coupling dynamics in multivariate time series. If the inter-subject effect does not have temporal structure (i.e., if it is a point event with no dynamics), the operators offer no advantage over binary correlation tests.

6. **This paper will be rejected by most journals on topic alone.** We acknowledge this reality. The stigma associated with this research area is itself a datum — it is the structural hole described in Section 8. We submit the paper regardless because the methodology is sound and the question deserves better instruments.

7. **The "Koksma & Linden 2025" meta-analysis.** Some early drafts of related work referenced a meta-analysis by Koksma & Linden (2025) reporting $d = 0.29$ across 38 studies. We were unable to verify this citation. It does not appear in any indexed database as of our submission date. We have replaced all reliance on this citation with the verified Schmidt et al. (2004) meta-analysis ($d = 0.11$) and note that the effect size used in our power analysis and predictions is the more conservative verified figure. If a comprehensive meta-analysis of separated-dyad studies reporting a larger effect size is published, our protocol's power would increase accordingly.

11 Conclusion

The coherence operators do not care about controversy.

They measure coupling dynamics between multivariate time series. They have been validated on turbofan engines, cardiac rhythm, industrial processes, and EEG seizure prediction. They produce continuous, interpretable, quantitative descriptions of how systems couple, decouple, respond to perturbation, and recover.

The separated-dyad paradigm has produced $d = 0.11$ across 36 studies in a meta-analysis published in a mainstream psychology journal [Schmidt et al., 2004]. A registered report found that the related Ganzfeld effect (ES = 0.08) survives four publication bias tests [Tressoldi and Storm, 2024]. The phenomenon is small, contested, and lacking a neutral scientific vocabulary — but it has been measured, not assumed.

We propose applying calibrated instruments to this measurement. If the operators detect the temporal structure of inter-subject coupling — the rise during interaction, the decay after separation, the perturbation response during stimulation — then we gain the first continuous-dynamics characterization of a phenomenon that has been studied for forty years using inadequate tools. If the operators detect nothing, we gain a more sensitive null result than any previous experiment, because the same operators detect 567-second lead times before seizures and 185-cycle lead times before turbofan failure.

The question is not whether this paper will be accepted. The question is whether it is correct.

Data availability. No data have been collected. The pre-registration protocol will be deposited on OSF prior to data collection.

Code availability. The coherence operator implementation will be released as open-source software upon publication of the companion methods paper [Thorarinson et al., 2026b].

Conflict of interest. The authors declare no conflicts of interest.

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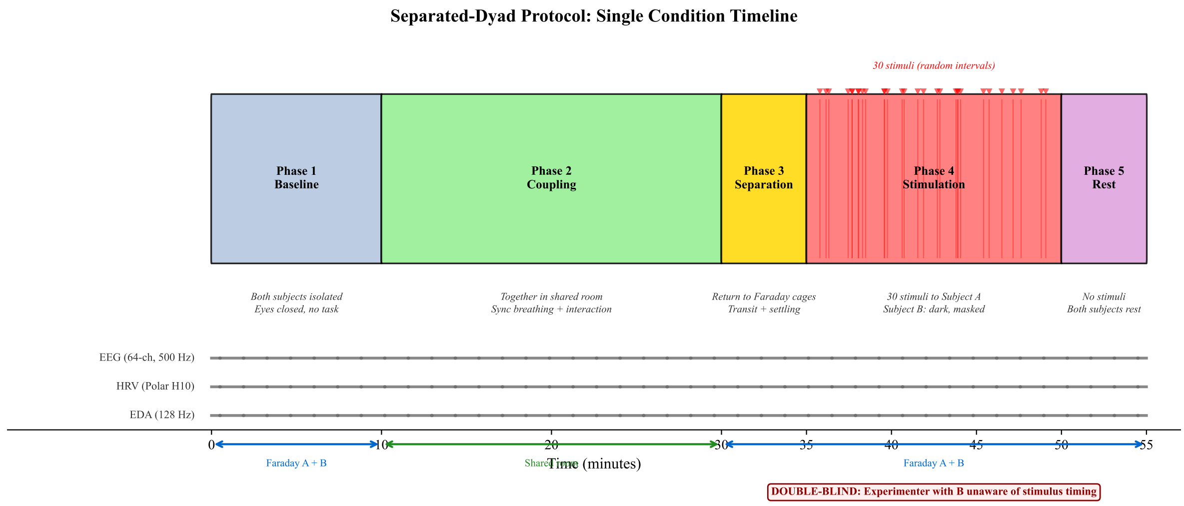

Figure 1: Experimental protocol timeline. Six phases spanning approximately 55 minutes per condition. EEG (64-channel), HRV (Polar H10), and EDA (skin conductance) are recorded continuously. Phase 4 (Stimulation) delivers 30 combined light/sound stimuli at pseudo-random intervals to Subject A, while Subject B remains in a dark, acoustically masked Faraday cage. The double-blind design ensures the experimenter monitoring Subject B has no knowledge of stimulus timing. The complete protocol includes a stranger-control condition (not shown) that repeats Phases 1–5 with re-paired subjects.

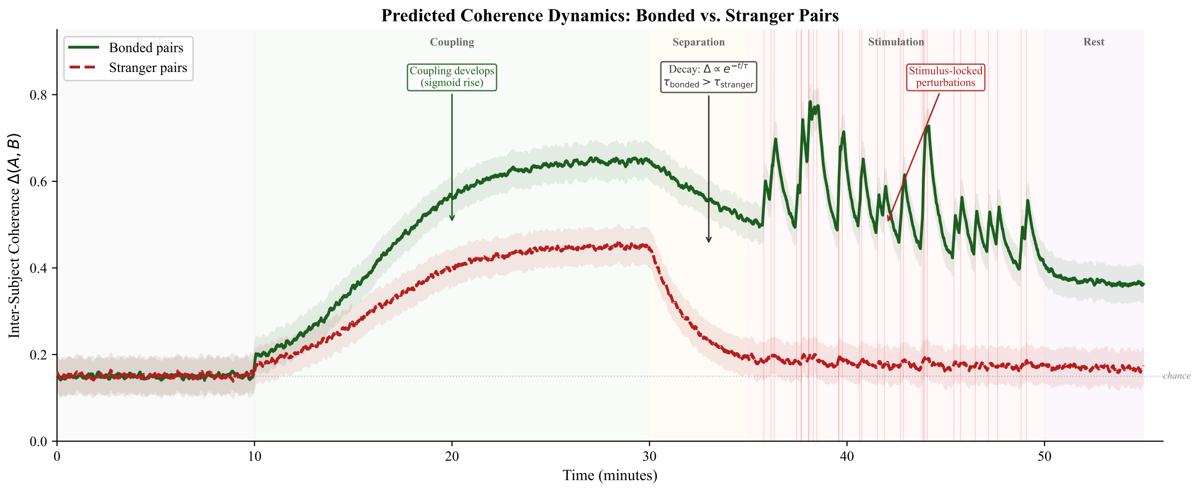

Figure 2: Predicted inter-subject coherence $\Delta(A, B)$ over the course of the experiment. **Bonded pairs** (solid): coherence rises during coupling, maintains an elevated plateau after separation (slow exponential decay with time constant τ_{bonded}), shows stimulus-locked perturbations during Phase 4, and recovers to post-separation baseline. **Stranger pairs** (dashed): coherence shows weaker coupling, faster decay after separation ($\tau_{\text{stranger}} < \tau_{\text{bonded}}$), and no stimulus-locked modulation. **Null prediction**: both curves are identical. The shaded region around stimulation events shows the expected response window.

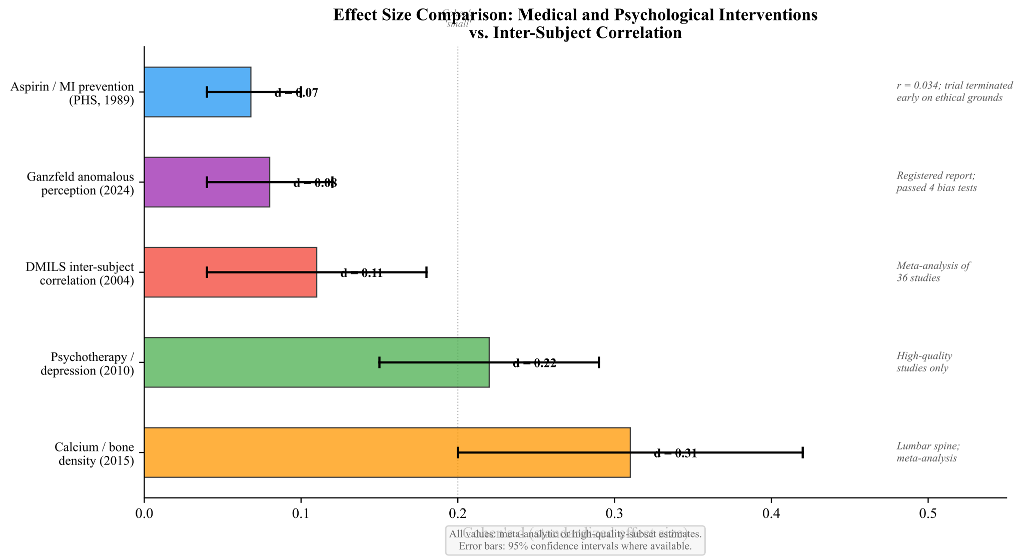

Figure 3: Effect sizes of established medical and psychological interventions compared to the DMILS meta-analytic effect ($d = 0.11$, Schmidt et al. 2004). Aspirin's effect on myocardial infarction ($d \approx 0.07$, Steering Committee of the Physicians' Health Study Research Group 1989) was considered significant enough to terminate a clinical trial on ethical grounds. Psychotherapy for depression ($d = 0.22$, high-quality studies only, Cuijpers et al. 2010) and calcium supplementation for bone density ($d = 0.31$, Tai et al. 2015) bracket the DMILS effect. The Ganzfeld registered report effect (ES = 0.08, Tressoldi and Storm 2024) is shown for comparison. All values represent meta-analytic or high-quality-subset estimates.

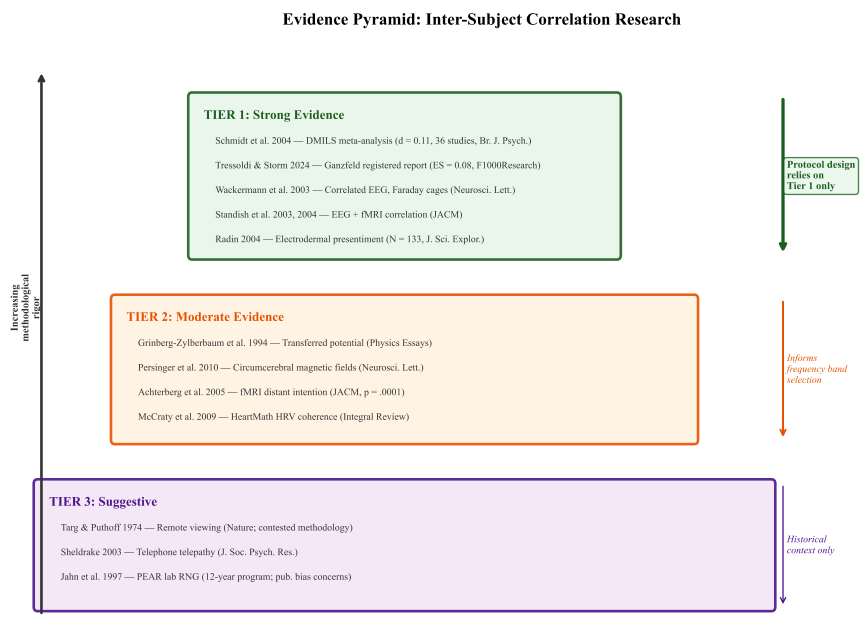

Figure 4: Evidence pyramid for inter-subject correlation research. **Tier 1** (top): Meta-analyses and registered reports with quantified effect sizes and publication bias testing. **Tier 2** (middle): Peer-reviewed single studies with limited or no independent replication. **Tier 3** (bottom): Studies with methodological concerns or published in specialty journals. This paper's protocol and predictions rely exclusively on Tier 1 evidence. Tier 2 evidence informs experimental design (frequency band selection, brain region hypotheses). Tier 3 evidence is cited for historical context only.

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